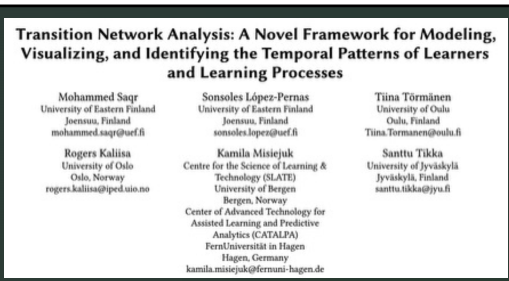
From Data toTheory

Building Explanatory Models in Educational Technology Research

1

1. Motivation and background

2



Transition Network Analysis: A Novel Framework for Modeling, Visualizing, and Identifying the Temporal Patterns of Learners and Learning Processes

Mohammed Saqr
University of Eastern Finland
Joensuu, Finland
mohammed.saqr@uef.fi

Sonsoles López-Pernas
University of Eastern Finland
Joensuu, Finland
sonsoles.lopez@uef.fi

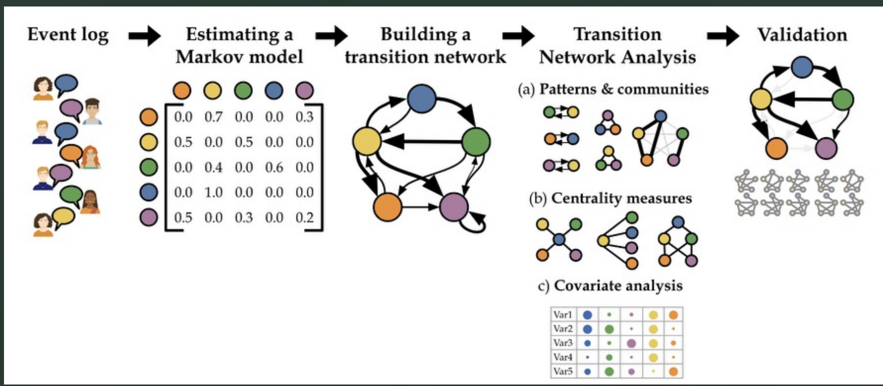
Tiina Törmänen
University of Oulu
Oulu, Finland
Tiina.Tormanen@oulu.fi

Rogers Kaliisa
University of Oslo
Oslo, Norway
rogers.kaliisa@iped.uio.no

Kamila Misiejuk
Centre for the Science of Learning & Technology (SLATE)
University of Bergen
Bergen, Norway
Center of Advanced Technology for Assisted Learning and Predictive Analytics (CATALPA)
FernUniversität in Hagen
Hagen, Germany
kamila.misiejuk@fernuni-hagen.de

Santtu Tikka
University of Jyväskylä
Jyväskylä, Finland
santtu.tikka@jyu.fi

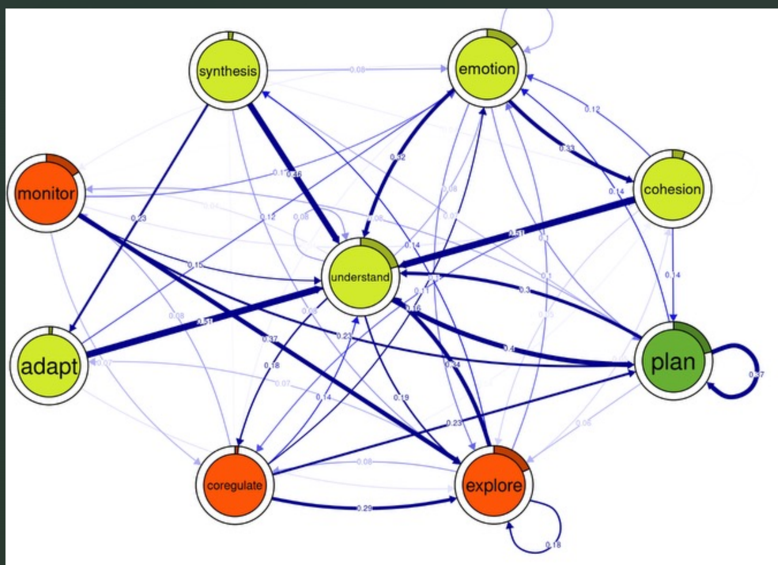
Example for sequence analysis



3

What phenomenon does this figure describe?

Does the figure explain the phenomenon?

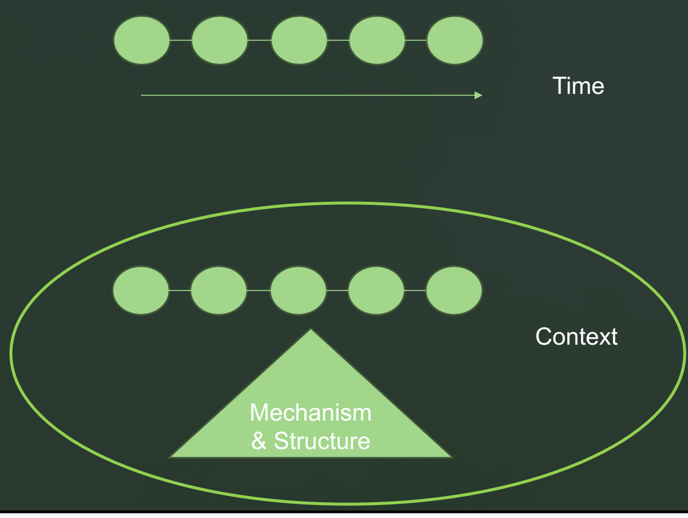


4

Flat vs layered ontology

Flat:

Layered:



5

From patterns to explanatory mechanisms

The pattern :
planning → confusion → monitoring → negotiation → revision → convergence

What produced this pattern of *productive recovery*?

How do we know that we now understand this phenomenon?

6

Scientific understanding = counterfactual inferential ability

Understanding = The ability to make correct inferences about what would happen if things were different – to answer "what if" questions.

- What would have happened if the group had not had a shared representation?
- What would have happened if the task had been more structured?
- What would have happened if the tool had allowed comments but not restructuring?
- Etc.

7

Mechanistic explanations

A mechanistic explanation explains a phenomenon by specifying the entities, activities, interactions, and contextual conditions that generate it.

confusion → monitoring → negotiation → revision

What mechanism(s) produced *productive recovery*?

"Retroduction"

A candidate mechanism is … and ..

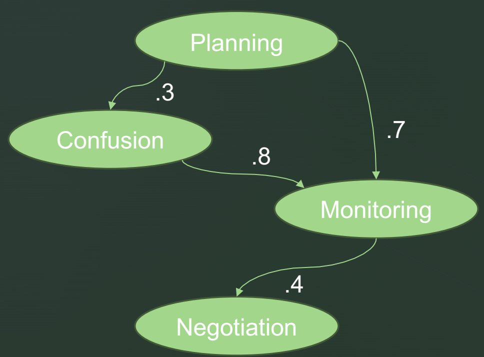
8

confusion → monitoring → negotiation → revision

A visible mismatch in a shared representation creates a regulatory opportunity. One student notices the mismatch. Because the mismatch is located in the shared object rather than attributed personally to another student, it becomes socially safer to challenge. The challenge prompts the group to articulate criteria for correctness. This shifts regulation from the individual to the group level. The group revises the representation, and the revised object changes what the group can notice and do next.

9

The flat ontology problem

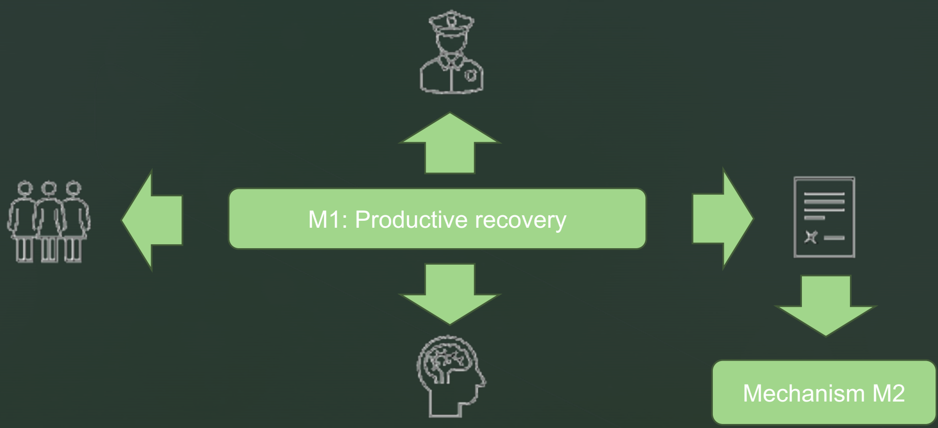


Sequence ≠ Mechanism

What would have to be true about learners, groups, tasks, tools, representations, and context for this sequence to be generated?

11

Layered mechanisms



12

Affordances of external representations

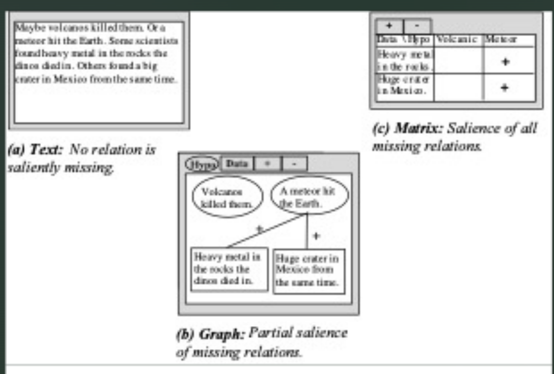


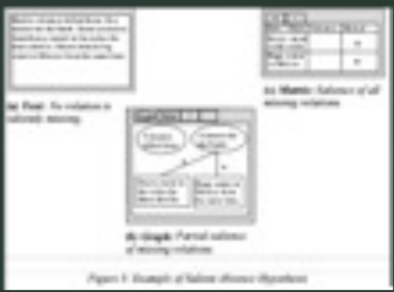
Figure 3: Example of Salient Absence Hypothesis

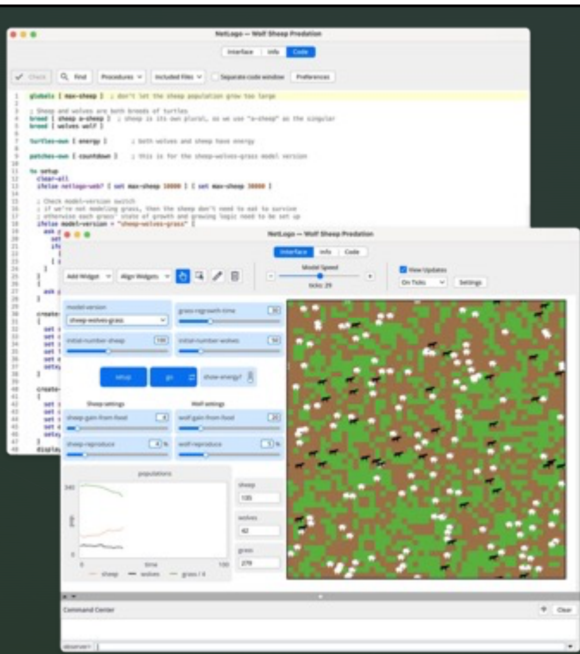
Different representational notations afford different kinds of inferences.

Suthers, D. D. (2001). Towards a systematic study of representational guidance for collaborative learning discourse. *Journal of Universal Computer Science*, *7*(3).

13

CSCL researchers need external representations, too.





By Uri Wilensky and the CCL lab - www.netlogo.org, GFDL, https://en.wikipedia.org/w/index.php?curid=80587273

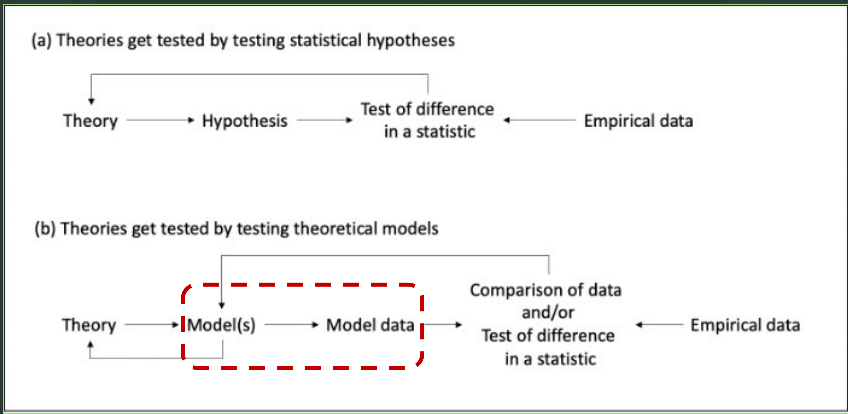
14

Descriptive data models versus generative mechanism models

- **Descriptive model:**
  - Identifies patterns in data
  - Represents event sequences
  - Data-dependent
  - Predictive
- **Generative model:**
  - Provides a possible process for generating the pattern
  - Represents assumptions about the production of event sequences
  - Theory-dependent
  - Predictive & explorative (counterfactual)

15

Extending our method toolbox with simulated data

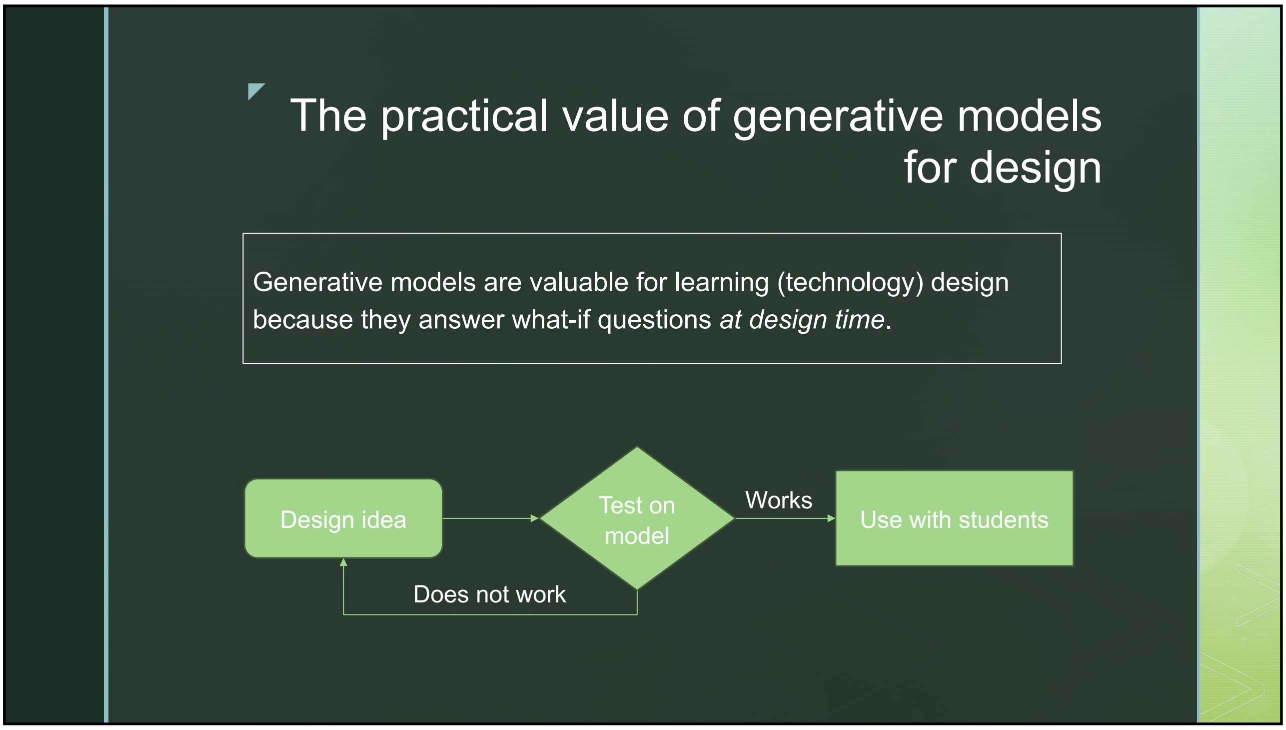


16

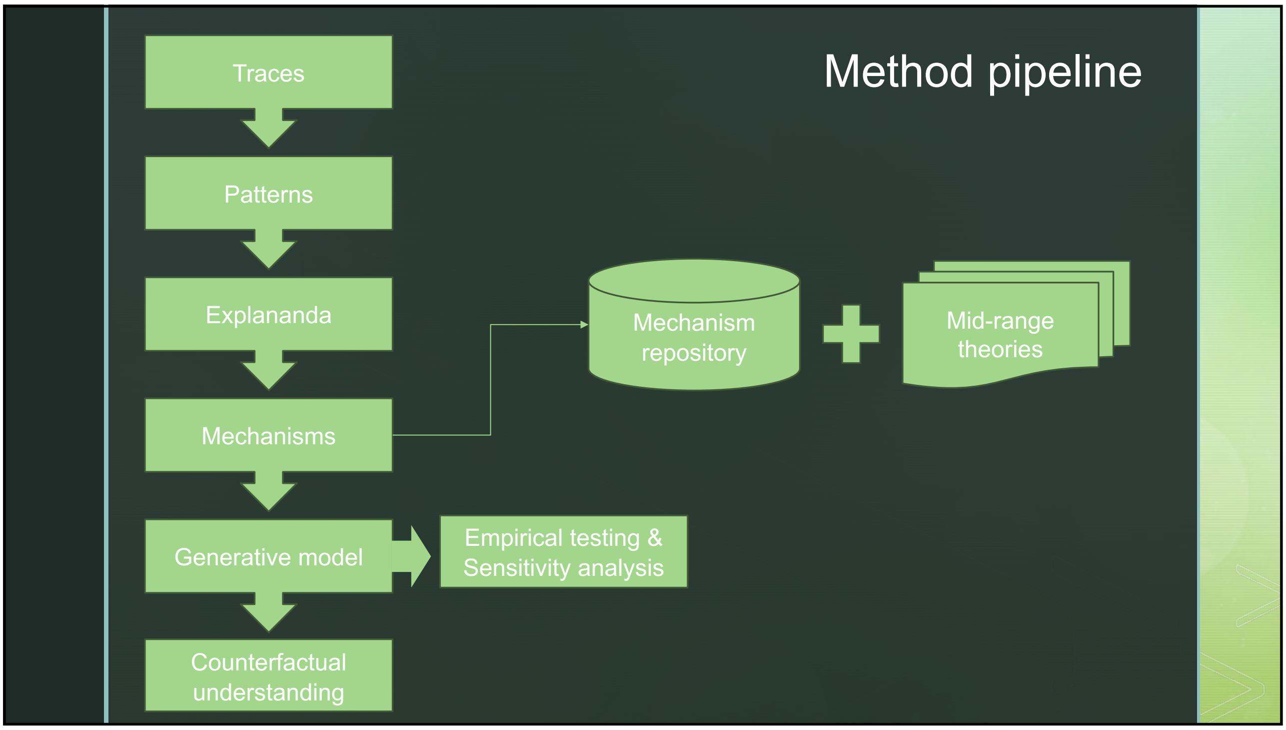
How can models be of scientific value even though they are wrong?

- Models are always simplifications and abstractions:
  - Missing elements (error of omission);
  - Elements that are not validated (error of commission).
- Models are never fully corresponding to empirical data.
- They are nevertheless valuable because
  - They provide a baseline: This much can be explained.
  - They allow for exploring the implications of theoretical assumptions.

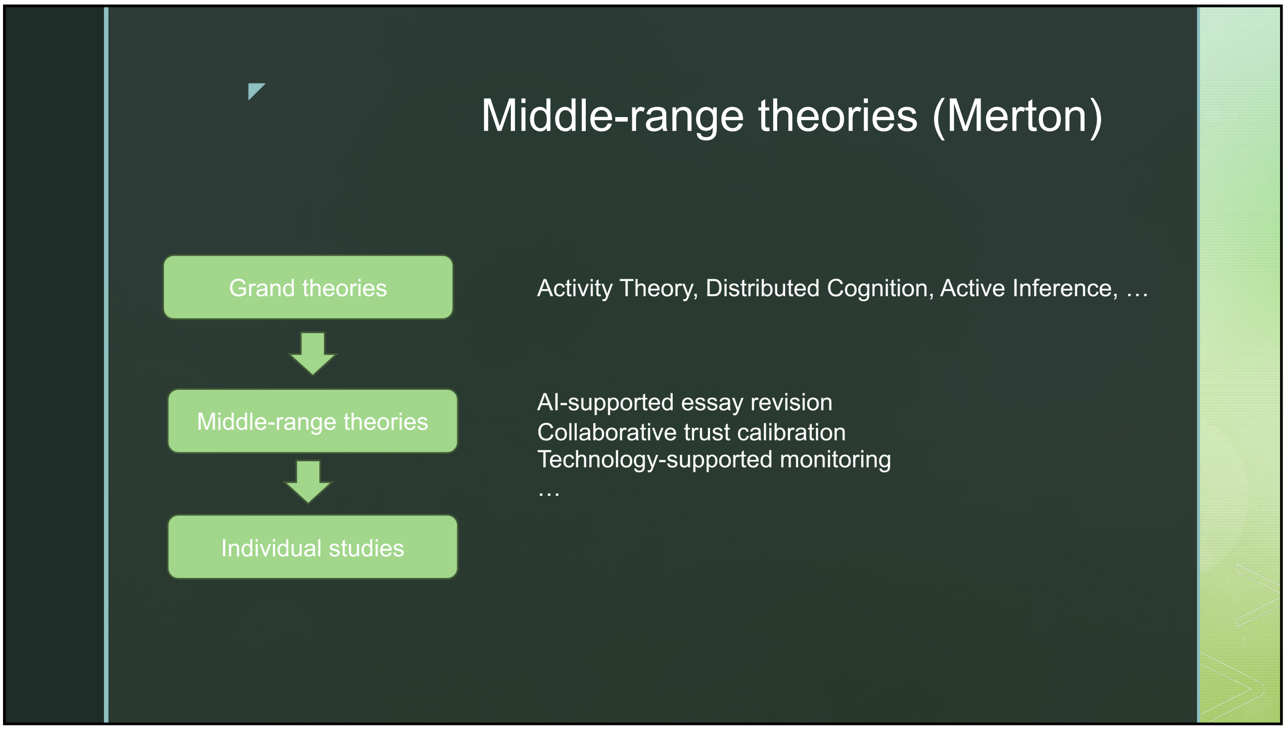
17


The practical value of generative models for design
Generative models are valuable for learning (technology) design because they answer what-if questions *at design time*.
Design idea
Test on model
Works
Use with students
Does not work

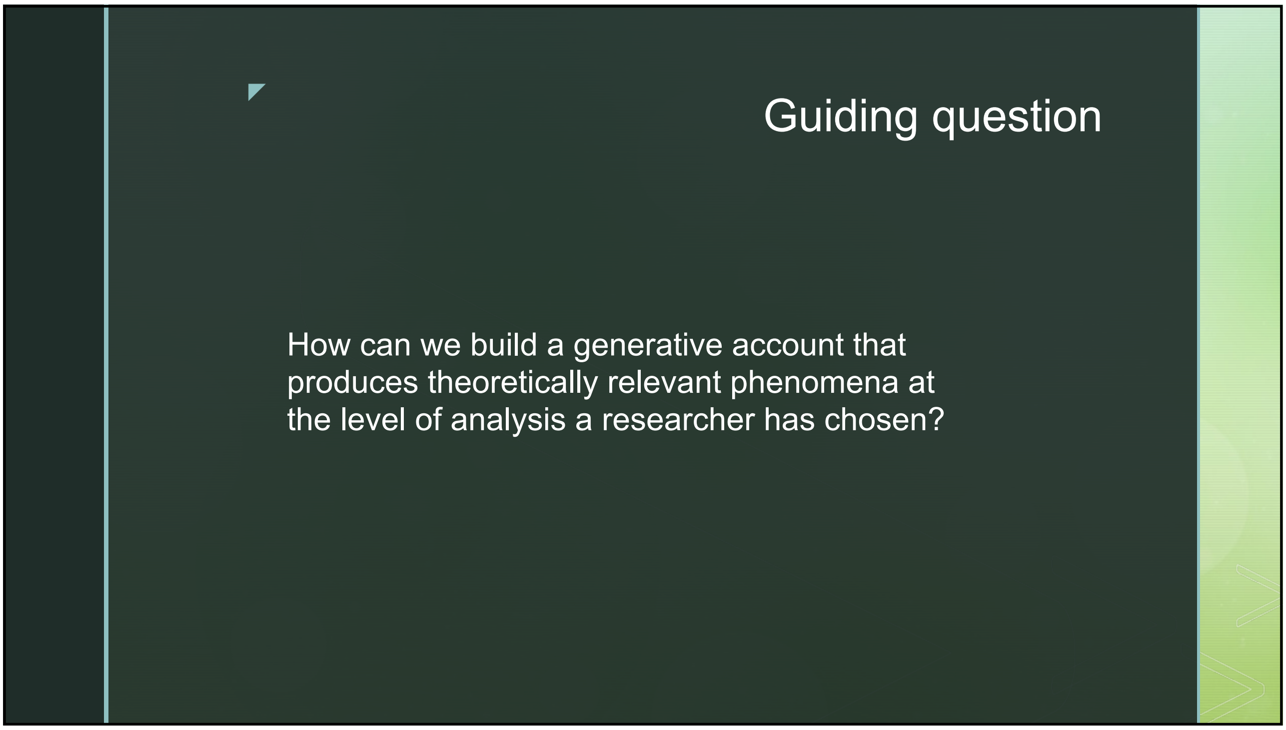
18


Method pipeline
Traces
Patterns
Explananda
Mechanisms
Generative model
Counterfactual understanding
Mechanism repository
Mid-range theories
Empirical testing & Sensitivity analysis

19


Middle-range theories (Merton)
Grand theories
Middle-range theories
Individual studies
Activity Theory, Distributed Cognition, Active Inference, …
AI-supported essay revision
Collaborative trust calibration
Technology-supported monitoring
…

20


Guiding question
How can we build a generative account that produces theoretically relevant phenomena at the level of analysis a researcher has chosen?

21